

Geometry Lead: Job Description for HR and Recruiters

CAM/AUTO | Mumbai | 8 October 2026

Also known as: Senior Software Engineer, CAD/CAM Geometry; CAD Kernel Developer; Computational Geometry Engineer

Experience	6 to 10 years writing software, at least 3 of them in CAD, CAM, CAD kernels, CNC simulation or robotics path planning
Salary	INR 40 to 50 lakh a year, plus a completion bonus at month 5. Band is an estimate; please advise if the market says otherwise
Location	Mumbai, on site preferred. Remote is possible for a strong candidate, with one week a month at the shop
Type	Full time, permanent. Notice periods up to 90 days are acceptable
Priority	Highest. This is the hire the whole plan depends on. Retained search approved
Reports to	Founder

About the company

A precision CNC machine shop in Mumbai is building its own software, CAM/AUTO. The software reads a customer's 3D CAD model and drawing and writes the program that a CNC milling machine runs to cut the part. The app already works (79 releases, tested on the shop floor). We are hiring a small team to make it accurate enough for daily production and to add a feature that learns from every part cut. Three roles, one founder, one senior CNC programmer from the shop, and an AI coding assistant (Claude Code) on every desk.

The job in plain words

This person is the chief engineer for the "geometry" part of the software: the code that reads the 3D model exactly and calculates the path the cutting tool follows. Today that code uses an approximation; this person replaces it with an exact method. It is a hands-on role: they write code every day, decide the technical approach, and check the work of one other developer and of the AI coding assistant. Success is measured on the machine: 27 of 30 real parts must come out right the first time.

What they will do day to day

- Read 3D CAD models (STEP files) exactly using a CAD kernel library (OpenCascade) and detect holes, pockets, walls and fillets with their true sizes.
- Calculate exact cutter paths with true arcs on every wall, to 0.01 mm.
- Build an independent check that re-reads the final machine program and catches any cut into the part or collision.
- Write automated tests that prove every number against the CAD file.
- Review all code that touches geometry, including code drafted by the AI assistant.
- Spend time at the machine when test parts are cut, and fix every defect to its root cause.

Must-have skills (screen on these)

Skill	What it means in practice	Weight
CAD kernel programming	Has written code against OpenCascade (OCCT). Parasolid, ACIS or CGM also accepted. This means programming with the library, not using CAD software like SolidWorks or Fusion 360 as a designer	Non-negotiable
Computational geometry	Has implemented or debugged polygon offsetting, Boolean operations or arc fitting (Clipper2, CGAL or their own code)	Non-negotiable
Programming languages	Strong in TypeScript or JavaScript, or strong in C++ and willing to work in TypeScript. Must read both	Non-negotiable
Testing numerical code	Writes automated tests for geometry and checks results against an independent source	Non-negotiable
Git and code review	Daily use of Git and GitHub pull requests	Required

Good to have (not required)

- CAM toolpath generation: pocketing, contouring, helical drilling, roughing strategies.
- G-code for Fanuc, Haas or Siemens CNC controllers.
- Material-removal simulation (dixel or voxel), collision checking.
- WebAssembly / Emscripten (compiling C++ to run in a browser).
- Has used a commercial CAM package: Fusion 360, Mastercam, NX CAM, hyperMILL, SolidCAM, PowerMill.
- Contributions to open-source geometry projects: OCCT, FreeCAD, CadQuery, OpenCAMLib, Clipper.

Not a match (please do not send)

- CAD designers, draughtsmen or CAM programmers who use CAD/CAM software but do not write software.

- Web or mobile app developers with no geometry background.
- 3D graphics or game developers (Unity, Unreal, three.js) with no CAD kernel experience.
- Data scientists and machine learning engineers. This role has no ML.

Where to find candidates

Companies	Autodesk India (Fusion 360, Inventor, PowerMill), Siemens Digital Industries Software (NX, Parasolid), Dassault Systemes (SolidWorks, CATIA), PTC (Creo), Hexagon Manufacturing Intelligence (Edgecam, NCSIMUL), HCLTech CAMWorks (formerly Geometric Ltd), Sandvik / CGTech (Vericut), Zeus Numerix, CAD/CAM start-ups
Institutions	IIT Bombay, IIT Madras, IIT Delhi and IISc computational geometry and CAD groups; PhD or master's graduates with industry experience
Job titles to search	Geometry Software Engineer, CAD Kernel Developer, CAM Developer, Computational Geometry Engineer, Geometric Modeling Engineer, Toolpath Developer, Senior Software Engineer CAD/CAM, Simulation Software Engineer

Screening questions HR can ask

Question	Answer to listen for
Which CAD kernel have you written code against?	OpenCascade / OCCT, Parasolid, ACIS or CGM. If they only name CAD software (SolidWorks, AutoCAD, Fusion), not a match
Have you implemented contour or polygon offsetting with arcs?	Yes, with details: Clipper, CGAL or own implementation; mentions self-intersections or arc fitting
Which languages do you write in daily?	C++ and/or TypeScript or JavaScript
Have you worked with CNC toolpaths or G-code?	Good to have; not a knock-out
Are you open to a paid one-day practical test?	Must say yes

Hiring process

1. Short call with the founder (30 minutes).
2. Paid one-day practical test on real material. The candidate is paid for the day whether or not we proceed.
3. Two-hour review of the test at the shop, in front of the machine.
4. Reference calls, then an offer within a week.

Keywords

Must-have keywords (resume should show most of these)

OpenCascade, OCCT, Parasolid, ACIS, CGM, CAD kernel, B-rep, STEP, computational geometry, polygon offsetting, Clipper2, CGAL, arc fitting, C++, TypeScript, JavaScript

Good-to-have keywords

CAM, toolpath, 2.5D, pocketing, G-code, Fanuc, Haas, Siemens Sinumerik, cutter compensation, dextral, voxel, material removal simulation, collision detection, WebAssembly, Emscripten, three.js, CMM, GD&T, Fusion 360, Mastercam, NX CAM, hyperMILL, PowerMill, OpenCAMLlib, FreeCAD, CadQuery

Boolean search string (LinkedIn Recruiter, Naukri)

("OpenCascade" OR "OCCT" OR "Parasolid" OR "ACIS" OR "CAD kernel" OR "B-rep" OR "computational geometry") AND ("CAM" OR "toolpath" OR "offsetting" OR "geometry") AND ("C++" OR "TypeScript" OR "JavaScript")

Titles and profiles to skip

SolidWorks designer, AutoCAD draughtsman, CAD operator, CNC programmer (no software), data scientist, machine learning engineer, Unity developer